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SARDAR PATEL UNIVERSITY

Master of Science in CA & IT (MSc-CA & IT)

Semester – II External Examinations (Regular), September 2020

PS02CIIT33 – Artificial Intelligence

Wednesday, 23rd September, 2020

Time: 03:00 p.m. to 05:00 p.m.

Max Marks: 70

Q1.A Choose the most appropriate option for each question.

[10]

- _____ largely includes rule of thumb also called heuristics.
A) Public knowledge B) private knowledge
C) Expert knowledge D) None of these
- The test in which the computer mimics a human and tries to fool other humans is called _____.
A) chinese room test B) turing test
C) AI test D) None of these
- Perception falls into category of _____ task.
A) mundane B) formal
C) expert D) executive
- _____ is general knowledge about a domain that an average person is expected to know.
A) Heuristics knowledge B) Commonsense knowledge
C) Domain knowledge D) Meta knowledge
- Qualitative expert knowledge is also known as _____ knowledge is acquired with protocol analysis.
A) Tacit B) Factual
C) Declarative D) Formal
- Factual knowledge uses _____ symbol for implies.
A) $\sim$ B) $\vee$
C) $\wedge$ D) $\Rightarrow$
- _____ maps elements of a fuzzy set to real numbered values in the interval 0 to 1.
A) Fuzzy logic B) Bi-valued logic
C) Membership function D) Genetic function
- In ANN, activation output ($\sum X_i W_i$) is measured with a predefined output, if it is greater than threshold value, output is ____ otherwise output is ____.
A) 0,0 B) 0,1
C) 1,0 D) 1,1
- _____ is a collection of such genes representing possible solutions in domain space.
A) Genetic cycle B) Genetic algorithm
C) Genotype D) Mutation
- _____ is an incremental change to make each member of population unique and enables new features to be introduced into the population.
A) Selection B) Mutation
C) Crossover D) Inversion

Q2 Fill in the Blanks and True or False

[8]

1. Medical diagnosis is a _____ task.
2. Public knowledge includes published definitions, facts and theories available in books, reports, research papers and so on. (True or False)
3. _____ is a component of knowledge that represents set of raw observation, alphabet, symbols or statements.
4. To represent Universal Quantification $\forall$ symbol is used to represent Factual Knowledge. (True or False)
5. _____ converts the fuzzy value into a crisp value.
6. The meaning of conjunction is given by an intersection function. (True or False)
7. A _____ is a template, defined over the alphabet $\{0,1,*\}$, which describes a pattern of bit strings in the search space $\{0,1\}^{\text{length}}$.
8. The degree of weakness is referred to the strength of genes in an individual. (True or False)

Q3. Answer in short (Attempt any 10 out of 12)

[20]

1. Differentiate between data and information.
2. List of characteristics of knowledge.
3. Write goal of artificial intelligence.
4. Draw the general structure of KBS.
5. List steps of knowledge acquisition.
6. List users of KBS.
7. What are fuzzy logic and crisp logic?
8. Explain linguistic variables.
9. What is Artificial Neural Networks (ANN)?
10. Define chromosomes and genotype.
11. What is stopping criteria in function optimization?
12. What is mutation?

Q4. Answer in detail (Attempt any 4 out of 8)

[32]

1. Write a detailed note on inference engine.
2. List types of knowledge and explain any three in detail.
3. Draw and explain Knowledge Based System Development Model.
4. Discuss Semantic Networks in detail.
5. Mention the role of membership function in fuzzy logic.
6. Explain Perceptrons in detail.
7. Write a detailed note on crossover operation.
8. Discuss in depth, the working of edge recombination technique.

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